A Minimal Uniformly Recurrent Subshift with a Periodic-Orbit Vacuum

Route-A structural certificate C144

Abstract

The two-sided Thue–Morse substitution subshift is nonempty, minimal, and uniformly recurrent, yet it has no shift-periodic point. We give a self-contained proof: for every proposed period p, an odd-popcount multiple $d = p(2^k - 1)$ forces a mismatch inside every sufficiently long Thue–Morse window. Thus every positive Artin–Mazur fixed-point count and primitive-cycle count vanishes and the source zeta is 1. Circulated finite substitution words are audited as periodic controls, with exact local and macroscopic defect ledgers; they are not points of the limiting subshift. This proves that rich recurrence alone need not supply a Route-A primitive-orbit layer.

1 Frozen language subshift

Let $t_n \in \{0, 1\}$ be the parity of the binary digit sum of $n \geq 0$. Equivalently, $t = 01101001\cdots$ is fixed by the substitution $0 \mapsto 01$, $1 \mapsto 10$. Write $w_q = t_0 \cdots t_{2^q-1}$ and let $\overline{w}_q$ denote bitwise complement. Freeze

$$X_{\mathrm{TM}} = \{x \in \{0, 1\}^{\mathbb{Z}} : \text{every finite factor of } x \text{ occurs in } t\}$$

with one left-shift iterate as the clock. The Artin–Mazur convention is

$$\zeta_{\mathrm{TM}}(z) = \exp\left(\sum_{n\geq 1} \#\,\mathrm{Fix}(\sigma^n|X_{\mathrm{TM}})\frac{z^n}{n}\right). \tag{1}$$

The language definition makes X_{TM} closed and shift invariant.

Binary concatenation without carries gives, for $0 \leq r < 2^q$,

$$t_{j2^q+r} = t_j \ \mathsf{xor}\ t_r. \tag{2}$$

Hence each aligned q-block is w_q or $\overline{w}_q$. Moreover $t_{2m} = t_m$ and $t_{2m+1} = 1 - t_m$, so every pair beginning at an even coordinate is complementary and neither 000 nor 111 occurs.

2 Uniform recurrence and minimality

Fix a factor u of t and choose q so that u occurs in w_q. Every interval of length $4 \cdot 2^q$ contains three complete aligned q-blocks. Their types are three consecutive symbols of t, not all 1, so one is w_q and contains u. Thus every factor returns with bounded gaps.

For nonemptiness, extend t arbitrarily on negative coordinates and take a product-topology limit of shifts whose origins tend to $+\infty$. Every fixed window of the shifted sequence eventually lies wholly in t, so every factor of the limit belongs to the language. If $x \in X_{\mathrm{TM}}$ and u has recurrence bound R, every length-R factor of x is a factor of t and therefore contains u. Consequently every orbit in X_{TM} meets every nonempty cylinder: X_{TM} is minimal.

3 All-period aperiodicity theorem

Fix a proposed positive period p. Choose an odd integer k strictly larger than the binary length of p and set

$$d = p(2^k - 1) = (p - 1)2^k + (2^k - p). \tag{3}$$

The low k bits of $2^k - p$ complement the k-bit expansion of $p - 1$. The two summands in (3) occupy disjoint binary blocks, whence

$$\mathrm{popcount}(d) = \mathrm{popcount}(p - 1) + k - \mathrm{popcount}(p - 1) = k. \tag{4}$$

Therefore d is a multiple of p and $t_d = 1 \neq t_0$.

Let b be the binary length of d, so $d < 2^b$. Every interval of t of length 2^{b+1} contains a complete b-aligned block. Equation (2) shows that its symbols at offsets 0 and d differ. Those offsets are congruent modulo p, so the interval is not p-periodic. If a p-periodic $x \in X_{\mathrm{TM}}$ existed, its window of that length would occur in t and would be p-periodic, a contradiction. Hence

$$\mathrm{Fix}(\sigma^n | X_{\mathrm{TM}}) = \varnothing \quad (n \geq 1), \qquad \zeta_{\mathrm{TM}}(z) = 1. \tag{5}$$

There are likewise no primitive shift cycles. This is not an absence of recurrence or invariant measures; it is specifically a periodic-orbit vacuum.

4 Periodic controls and validation

Circulate w_k to obtain $c_k = w_k^\infty$. Its least period is 2^k. Indeed, a least cyclic period divides 2^k; any proper divisor also divides 2^{k-1} and would make the two halves equal, whereas the second half of w_k complements the first. For a width-m window with $m \leq 2^k$, only the $m - 1$ seam-crossing starts could be extrinsic, so the invalid rooted fraction is at most $(m-1)/2^k$.

The exact ledger gives a sharper scale contrast. For each width m, choose $2^q \geq m$. Equation (2) shows that every intrinsic m-factor lies in one of $w_q w_q, w_q \overline{w}_q, \overline{w}_q w_q, \overline{w}_q \overline{w}_q$; the pairs $00, 01, 10, 11$ all occur already in 01101001. This makes the finite language membership tests exhaustive. At levels $2 \leq k \leq 12$, all 145 audited local cells through width 16 have zero invalid rooted windows. At levels $2 \leq k \leq 9$, every rooted window of width $2^{k+1} + 1$ is extrinsic. This latter statement is a finite checked control, not an all-k theorem, and neither finite cutoff enters the proof of (5). Equation (5) already shows that no periodic c_k belongs to X_{TM}.

The independent checker passes 172,437 assertions, SymPy passes 83 exact checks, isolated production is byte-identical, and all 37 hostile cases (36 repaired-hash plus one stale-hash) are rejected. The strict verdict is (`A1_FAIL`, `A2_FAIL`, `A3_FAIL`, `A4_FAIL`), overall `ROUTE_A_REJECTED`. We claim no target divisor, target functional equation or counting law, arithmetic/local factor, root number, automorphy, natural operator lift, Hilbert–Pólya operator, or Route-B authorization. The scope literal is `NO_BAD_EULER_OR_ROOT_NUMBER`.